

Lecture 5: Singular Value Decomposition

INF1004 Linear Algebra (Part II)

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Outline

- 1 Problem with Non-Square Matrices
- 2 Diagonalization of $A^T A$
- 3 Singular Value Decomposition of A
- 4 Using AA^T Instead of $A^T A$
- 5 Application of SVD in Recommender Systems

Recap: Eigendecomposition / Diagonalization

To obtain the eigendecomposition (or diagonalization) of a matrix, we put the matrix in the form:

$$A = PDP^{-1}$$

Where:

- $D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$ (Diagonal matrix of eigenvalues)
- $P = [v_1 \ v_2 \ \dots \ v_n]$ (Matrix of column eigenvectors)
- $P^{-1} = [v_1 \ v_2 \ \dots \ v_n]^{-1}$

Notice

Eigendecomposition only applies to **square matrices**, since eigenvalues depend on determinants.

Problem: What if Matrix A is Not a Square Matrix?

The Issue

A non-square matrix does not have a determinant, henceforth eliminating the existence of eigenvalues and eigenvectors.

So a non-square matrix cannot be diagonalized?

Diagonalization of $A^T A$

Although we cannot diagonalize a non-square matrix A , we can consider firstly diagonalizing $A^T A$, which can be loosely interpreted as something like A^2 , followed by taking the square root.

Because:

$$A_{(n \times m)}^T A_{(m \times n)} = (A^T A)_{(n \times n)}$$

So $A^T A$ is a square matrix, hence possibly diagonalizable. Of course, $A_{(m \times n)} A_{(n \times m)}^T = (A A^T)_{(m \times m)}$ is also square. Hence, we can also try to diagonalize $A A^T$.

Note: We shall choose the smaller square matrix to diagonalize for simplicity.

When Do We Choose $A^T A$ or AA^T ?

For an $m \times n$ matrix A , both products are square:

$$A^T A \in \mathbb{R}^{n \times n}, \quad AA^T \in \mathbb{R}^{m \times m}$$

- Use $A^T A$ when $n < m$ (fewer columns than rows): smaller eigenproblem.
- Use AA^T when $m < n$ (fewer rows than columns): smaller eigenproblem.
- If $m = n$, either choice works; pick the one that is cleaner computationally.

Practical rule: diagonalize the smaller of $A^T A$ and AA^T for efficiency and numerical convenience.

Factorization of Any Matrix A

Remember, diagonalization is just factorization of the matrix into 'nice' matrices multiplied together. The whole idea is:

- 1 Assume A can be factorized, like: $A = XYZ$, where X, Y, Z satisfy 'nice' properties.
- 2 Make use of diagonalization: $A^T A = PDP^{-1}$.
- 3 Combine the two steps to work out the matrices X, Y , & Z .

If this doesn't work out or X, Y, Z are not 'nice', scrap it and try another approach. It's free anyway.

Factorization of Any Matrix A (Derivation)

So, firstly, we assume A can be factorized into $A = U\Sigma V^T$, where U & V are **orthonormal matrices** (matrices that satisfy $O^T O = I$), and derive:

$$\begin{aligned} A^T A &= (U\Sigma V^T)^T (U\Sigma V^T) \\ &= (V\Sigma^T U^T)(U\Sigma V^T) \\ &= V\Sigma^T (U^T U) \Sigma V^T \\ &= V\Sigma^T I \Sigma V^T \\ &= V\Sigma^2 V^T \end{aligned}$$

And since $A^T A$ is square, it can also be diagonalized as $A^T A = PDP^{-1}$. So:

$$A^T A = V\Sigma^2 V^T = PDP^{-1}$$

Factorization of Any Matrix A (Combining)

Combine the above and we get:

$$A^T A = V[\Sigma^2]V^T = P[D]P^{-1}$$

From this comparison, we deduce:

- $V = P$: Matrix of orthonormal eigenvectors of $A^T A$.
- $\Sigma = \sqrt{D}$: Square root of diagonal matrix of eigenvalues of $A^T A$.
- $V^T = P^{-1}$: Only possible if eigenvectors are orthonormal.

Goal Achieved Partially

Remember our goal is to find 3 factors viz. U , Σ , and V from $A = U\Sigma V^T$. Notice that we obtained two of the three factors, V^T & Σ , just by finding eigenvalues and eigenvectors of $A^T A$.

Finding the Matrix U

Lastly, to find U in $A = U\Sigma V^T$, we multiply both sides by V :

$$AV = U\Sigma V^T V \Rightarrow AV = U\Sigma I \Rightarrow AV = U\Sigma$$

This can be expanded to give:

$$A \begin{bmatrix} | & | & \dots \\ v_1 & v_2 & \dots \\ | & | & \dots \end{bmatrix} = \begin{bmatrix} | & | & \dots \\ u_1 & u_2 & \dots \\ | & | & \dots \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots \\ 0 & \sigma_2 & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$
$$\begin{bmatrix} | & | & \dots \\ Av_1 & Av_2 & \dots \\ | & | & \dots \end{bmatrix} = \begin{bmatrix} | & | & \dots \\ \sigma_1 u_1 & \sigma_2 u_2 & \dots \\ | & | & \dots \end{bmatrix}$$

So, each vector u_i can be calculated from:

$$Av_i = \sigma_i u_i \Rightarrow \mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i$$

Singular Value Decomposition (SVD)

Finally, we defined all 'nice' terms in $A = U\Sigma V^T$. This is known as the Singular Value Decomposition (SVD) of an $m \times n$ matrix A .

$$A_{(m \times n)} = U_{(m \times m)} \Sigma_{(m \times n)} V_{(n \times n)}^T$$

- $U = [u_1 \quad u_2 \quad \dots \quad u_m]$ (Matrix of orthonormal vectors. $u_i = \frac{1}{\sigma_i} A v_i$)
- $\Sigma = \begin{bmatrix} \sqrt{\lambda_1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sqrt{\lambda_n} \\ 0 & 0 & 0 \end{bmatrix}$ (Matrix of rooted eigenvalues for $A^T A$. Pad rows/columns of zeros to get $m \times n$)
- $V^T = [v_1 \quad v_2 \quad \dots \quad v_n]^T$ (Transposed matrix of orthonormal eigenvectors for $A^T A$)

SVD Terminologies and The Most Powerful Factorization

$$A = U\Sigma V^T$$

- Each column vector, u_i , is called a **left singular vector**.
- Each scalar, σ_i , is called a **singular value**.
- Each row vector, v_i^T , is called a **right singular vector**.

The SVD of a matrix is not unique. Singular values σ_i are usually arranged in descending order.

The Most Powerful Factorization

The SVD is able to factorize **ANY** matrix, both real (& complex), into only diagonal/orthonormal (unitary) matrices.

Using AA^T Instead of $A^T A$

Start from the SVD form $A = U\Sigma V^T$:

$$AA^T = (U\Sigma V^T)(U\Sigma V^T)^T = U\Sigma V^T V \Sigma^T U^T = U(\Sigma \Sigma^T) U^T$$

- This is an eigendecomposition form: $AA^T = U\Lambda U^T$.
- So the eigenvectors of AA^T are the columns of U (left singular vectors).
- Eigenvalues are σ_i^2 (diagonal of $\Sigma \Sigma^T$).

Why $A^T A$ Gives V but AA^T Gives U

Compare both identities:

$$A^T A = V(\Sigma^T \Sigma) V^T, \quad AA^T = U(\Sigma \Sigma^T) U^T$$

- Diagonalizing $A^T A$ gives right singular vectors (V).
- Diagonalizing AA^T gives left singular vectors (U).
- After finding U from AA^T , recover V via

$$v_i = \frac{1}{\sigma_i} A^T u_i \quad (\sigma_i > 0).$$

Example: Evaluate the SVD

Problem: Evaluate the SVD of $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}$

A is a 3×2 matrix. Therefore:

- U will be a 3×3 matrix.
- Σ will be a 3×2 matrix.
- V will be a 2×2 matrix.

Step 1: Find $A^T A$

$$A^T A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+4+4 & 2+4+2 \\ 2+4+2 & 4+4+1 \end{bmatrix} = \begin{bmatrix} 9 & 8 \\ 8 & 9 \end{bmatrix}$$

Example: Find Eigenvalues (Σ) and Eigenvectors (V)

Step 2: Find eigenvalues λ for $A^T A$

$$|A^T A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 9 - \lambda & 8 \\ 8 & 9 - \lambda \end{vmatrix} = 0$$

$$(9 - \lambda)(9 - \lambda) - 64 = 0 \Rightarrow \lambda^2 - 18\lambda + 17 = 0 \Rightarrow (\lambda - 17)(\lambda - 1) = 0$$

$$\lambda_1 = 17, \lambda_2 = 1. \quad \text{Therefore, } \Sigma = \begin{bmatrix} \sqrt{17} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Step 3: Find eigenvectors $\vec{x}_1, \vec{x}_2$

- For $\lambda_1 = 17$: $\begin{bmatrix} -8 & 8 \\ 8 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow -8x_1 + 8x_2 = 0 \Rightarrow x_1 = x_2. \quad \vec{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- For $\lambda_2 = 1$: $\begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow 8x_1 + 8x_2 = 0 \Rightarrow x_1 = -x_2. \quad \vec{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Example: Normalizing V

Step 4: Normalize $\vec{x}_1$ and $\vec{x}_2$ to find V

Magnitude $|\vec{x}_1| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

$$v_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)^T$$

Magnitude $|\vec{x}_2| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$.

$$v_2 = \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}} \right)^T$$

$$\text{So, } V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \text{ and } V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

(Please make sure you do the normalization, without which you will not get $U\Sigma V^T = A$)

Example: Finding U

Step 5: Find U using $u_i = \frac{1}{\sigma_i}Av_i$

$$u_1 = \frac{1}{\sigma_1}Av_1 = \frac{1}{\sqrt{17}} \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{17}} \begin{bmatrix} \frac{3}{\sqrt{2}} \\ \frac{4}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{34}} \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

$$u_2 = \frac{1}{\sigma_2}Av_2 = \frac{1}{1} \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Example: Finding u_3

Since U is 3×3 , we need a third column u_3 . All u_i must be mutually perpendicular. We can use the cross product of u_1 and u_2 :

$$\vec{u}_3^* = \vec{u}_1 \times \vec{u}_2 = \frac{1}{\sqrt{68}} \left(\begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix} \times \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right) = \frac{1}{\sqrt{68}} \begin{bmatrix} 4 \\ -6 \\ 4 \end{bmatrix} = \frac{1}{\sqrt{17}} \begin{bmatrix} 2 \\ -3 \\ 2 \end{bmatrix}$$

Normalizing $\vec{u}_3^*$ gives:

$$u_3 = \begin{bmatrix} \frac{2}{\sqrt{17}} \\ \frac{-3}{\sqrt{17}} \\ \frac{2}{\sqrt{17}} \end{bmatrix}$$

Example: Final $U\Sigma V^T$

$$\text{So, } U = \begin{bmatrix} \frac{3}{\sqrt{34}} & \frac{-1}{\sqrt{2}} & \frac{2}{\sqrt{17}} \\ \frac{4}{\sqrt{34}} & 0 & \frac{-3}{\sqrt{17}} \\ \frac{3}{\sqrt{34}} & \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{17}} \end{bmatrix}$$

Therefore, $A = U\Sigma V^T =$

$$\begin{bmatrix} \frac{3}{\sqrt{34}} & \frac{-1}{\sqrt{2}} & \frac{2}{\sqrt{17}} \\ \frac{4}{\sqrt{34}} & 0 & \frac{-3}{\sqrt{17}} \\ \frac{3}{\sqrt{34}} & \frac{1}{\sqrt{2}} & \frac{2}{\sqrt{17}} \end{bmatrix} \begin{bmatrix} \sqrt{17} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

If you compute the right side, you will get exactly $\begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 2 & 1 \end{bmatrix}$, which is the matrix A. We found the nice matrices!

Verification & Uniqueness

Question: Is there any other value of U that satisfies $U\Sigma V^T = A$?

Answer: Yes. Try replacing the last column vector u_3 with $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 100 \\ 200 \\ 300 \end{bmatrix}$, or even $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

Because it gets multiplied by the row of zeros in Σ , they all compute $U\Sigma V^T$ to still be equal to A . However, for U to remain an *orthogonal matrix* ($U^T U = I$), we must pad it with orthonormal vectors from the null space.

SVD in Recommender Systems

An Application of SVD in Online Platforms (e.g., YouTube, Google Maps)

Example: Google Maps wants to recommend food places to users based on the ratings given by each user. Say, we have 2 users with the ratings below:

User	Laksa Mama	Prata Bala	Salad Go
Wong	5	4	0
Fit	0	0	5

By applying SVD to this dataset, we can discover '**concepts**' underlying the preferences of each user, hence enabling Maps to give better recommendations.

SVD in Recommender Systems (Computation)

Firstly, we denote the dataset in a matrix A :

$$A = \begin{bmatrix} 5 & 4 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Computing AA^T , we get:

$$AA^T = \begin{bmatrix} 41 & 0 \\ 0 & 25 \end{bmatrix}$$

So the eigenvalues are $\lambda_1 = 41$, $\lambda_2 = 25$ and

$$\sigma_1 = \sqrt{41} \approx 6.4, \quad \sigma_2 = 5.$$

Since AA^T is diagonal here, the left singular vectors are directly:

$$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Detailed Calculation of u and v (1/2)

From

$$AA^T = \begin{bmatrix} 41 & 0 \\ 0 & 25 \end{bmatrix}, \quad |AA^T - \lambda I| = (41 - \lambda)(25 - \lambda) = 0,$$

we get $\lambda_1 = 41, \lambda_2 = 25$ and $\sigma_1 = \sqrt{41}, \sigma_2 = 5$.

Solve for eigenvectors of AA^T :

$$(AA^T - 41I)u = 0 \Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & -16 \end{bmatrix} u = 0 \Rightarrow u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

$$(AA^T - 25I)u = 0 \Rightarrow \begin{bmatrix} 16 & 0 \\ 0 & 0 \end{bmatrix} u = 0 \Rightarrow u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Detailed Calculation of u and v (2/2)

Now compute $v_i = \frac{1}{\sigma_i} A^T u_i$ using

$$A^T = \begin{bmatrix} 5 & 0 \\ 4 & 0 \\ 0 & 5 \end{bmatrix}.$$

$$v_1 = \frac{1}{\sqrt{41}} A^T u_1 = \frac{1}{\sqrt{41}} \begin{bmatrix} 5 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{5}{\sqrt{41}} \\ \frac{4}{\sqrt{41}} \\ 0 \end{bmatrix},$$

$$v_2 = \frac{1}{5} A^T u_2 = \frac{1}{5} \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

SVD in Recommender Systems (Factorized)

Finally, we write down the SVD of A :

$$A = U\Sigma V^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{41} & 0 & 0 \\ 0 & 5 & 0 \end{bmatrix} \begin{bmatrix} \frac{5}{\sqrt{41}} & \frac{4}{\sqrt{41}} & 0 \\ 0 & 0 & 1 \\ -\frac{4}{\sqrt{41}} & \frac{5}{\sqrt{41}} & 0 \end{bmatrix}$$

Dimensionality Reduction (PCA)

Here A is 2×3 , but only two singular values are nonzero. In larger datasets, directions associated with very small singular values can be truncated to reduce dimensionality.

This concept is known as **dimensionality reduction** using Principal Component Analysis (PCA).

Interpretation of the SVD

An interpretation of the $A = U\Sigma V^T$ factors could be:

- **U : User-side factors.** Since AA^T is diagonal here, users are already aligned with the two dominant directions.
- **Σ : Strength of latent preference directions.** The first direction ($\sqrt{41} \approx 6.4$) is stronger than the second (5).
- **V^T : Place to Cuisine Correlation.** (Places in columns).
 - Laksa and Prata load on v_1 (local-delight direction).
 - Salad aligns with v_2 (a different preference direction).

Conclusion: SVD reveals hidden structures. Wong prefers Local Delights; Fit prefers a different direction captured mainly by Salad.

Summary

Once we understand the links between user, cuisine, and places, we can make intelligent recommendations to the users.

- The actual application in recommendation systems is usually an advanced form of matrix factorization (e.g., handling sparse matrices), but SVD forms the mathematical foundation.
- SVD can be used to discover **hidden structures** in data.
- SVD contributes significantly to **dimensionality reduction** (PCA).

Key Takeaway: Matrix factorization, such as eigendecomposition and SVD, are heavily used in practical real-world applications!

End of Linear Algebra. Have fun!